Bio-Plex
Manager™ software
recommended

Bio-Plex Pro™ Assays

Angiogenesis

Instruction Manual

For technical support, contact your local Bio-Rad office or in the US, call 1-800-4BIORAD (1-800-424-6723).

For research use only. Not for diagnostic procedures.

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Section 1
Introduction

Angiogenesis is a critical component of various normal and pathological processes. A regulated angiogenic process includes the growth of new blood vessels for tissue repair, fetal development and female reproductive cycle. An uncontrolled angiogenic growth contributes to a major pathological component of diseases such as cancer, cardiovascular disease, and rheumatoid arthritis. Many proteins involved in the uncontrolled angiogenic growth are candidate drug targets relevant for the development of cancer therapies. Developing these therapies involves measurement of various angiogenesis biomarkers.

Bio-Plex Pro™ angiogenesis assays are magnetic bead-based multiplex assays that allow the measurement of angiogenesis biomarkers in diverse matrices including serum, plasma, and cell/tissue culture supernatant. The multiplexing feature makes it possible to quantitate the level of multiple angiogenesis targets in a single well of a 96-well microplate in just 3 hours, using as little as 12.5 µl of serum or plasma, or 50 µl of culture supernatant.

As one of the most recent additions to the Bio-Plex® suspension array system, these assays incorporate magnetic beads into their design. The magnetic beads allow the use of an assay protocol similar to non-magnetic Bio-Plex assays, with the option of using magnetic separation of wash steps instead of vacuum filtration (and allows automation of many of the steps).

Bio-Plex Manager™ software is recommended for Bio-Plex Pro angiogenesis assays. Instructions for Luminex xPONENT software are also provided. For instructions using other xMAP system software packages, contact Bio-Rad Technical support or your Bio-Rad field application specialist.

For a current listing of Bio-Plex angiogenesis assays, visit us on the Web at **www.bio-rad.com/bio-plex/**

Section 2
Principle

Technology

The Bio-Plex® suspension array system is built around three core technologies. The first is a novel technology that uses up to 100 unique fluorescently dyed beads (xMAP technology) that permit the simultaneous detection of up to 100 different types of molecules in a single well of a 96-well microplate. The second is a flow cytometer with two lasers and associated optics to measure the different molecules bound to the surface of the beads. The third is a high-speed digital signal processor that efficiently manages the fluorescent output.

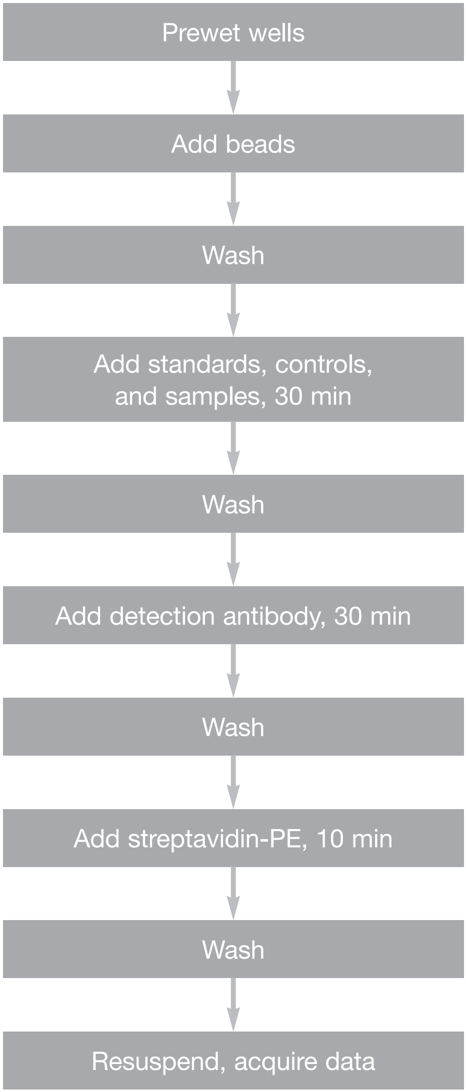
Assay Format

The principle of these 96-well plate-formatted, bead-based assays is similar to a capture sandwich immunoassay. An antibody directed against the desired angiogenesis target is covalently coupled to internally dyed beads. The coupled beads are allowed to react with a sample containing the angiogenesis target. After a series of washes to remove unbound protein, a biotinylated detection antibody specific for a different epitope is added to the reaction. The result is the formation of a sandwich of antibodies around the angiogenesis target. Streptavidin-phycoerythrin (streptavidin-PE) is then added to bind to the biotinylated detection antibodies on the bead surface.

Data Acquisition and Analysis

Data from the reaction are then acquired using the Bio-Plex suspension array system (or Luminex system), a dual-laser, flow-based microplate reader system. The contents of the well are drawn up into the reader. The lasers and associated optics detect the internal fluorescence of the individual dyed beads as well as the fluorescent signal on the bead surface. This identifies each assay and reports the level of target protein in the well. Intensity of fluorescence detected on the beads indicates the relative quantity of targeted molecules. A high-speed digital processor efficiently manages the data output, which is further analyzed and presented as fluorescence intensity on Bio-Plex Manager™ software.

Assay Workflow



Section 3
Bio-Plex® Reagent Kit
Product Description

Bio-Plex Pro™ angiogenesis assays require the use of the Bio-Plex® angiogenesis reagent kit to run the multiplex panel.

Component of the Bio-Plex Angiogenesis Reagent Kit	171-304060 1x96-Well Format	171-304061 10x96-Well Format
Bio-Plex assay buffer Store at 4ºC. Do not freeze.	1 x 75 ml	1 x 750 ml
Bio-Plex wash buffer Store at 4ºC. Do not freeze.	1 x 150 ml	2 x 750 ml
Bio-Plex detection antibody diluent Store at 4ºC. Do not freeze.	1 x 15 ml	1 x 150 ml
Streptavidin-PE (100x) Store at 4ºC. Do not freeze.	1 vial	1 vial
Sterile filter plate (96-well) with cover and tray	1 plate	10 plates
Sealing tape	1 pack of 4	10 packs of 4 (40)
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Storage and Stability

Kit components should be stored at 4ºC and never frozen. Coupled magnetic beads and streptavidin-PE should be stored in the dark. All components are guaranteed for up to 6 months from the date of purchase when stored as specified in this manual.

Section 4
Required Materials

Bio-Plex Pro™ angiogenesis assays require the Bio-Plex Pro™ angiogenesis reagent kit and a multiplex angiogenesis assay panel. The Bio-Plex Pro angiogenesis assay panels contain the following components.

- Antibody-conjugated beads (25x concentration)
- Detection antibody (10x concentration)
- Angiogenesis standard (2 vials, lyophilized)
- Angiogenesis control (2 vials, lyophilized)

Visit us on the web at **www.bio-rad.com/bio-plex/** for our latest list of assays.

Section 5
Recommended Materials

Item	Ordering Information
Bio-Plex human serum diluent kit For tissue culture samples, dilute samples and standards with appropriate tissue culture media	**Bio-Rad catalog #171-305000 (1x96)** **Bio-Rad catalog #171-305001 (10x96)**
Bio-Plex Pro human angiogenesis standards Additional standards sold separately (optional)	**Bio-Rad catalog #171-D40003 (2 vials, lyophilized)** **Bio-Rad catalog #171-D40004 (50 vials, lyophilized)**
Bio-Plex® suspension array system (or Luminex System)	**Bio-Rad catalog #171-000205**
Bio-Plex validation kit **Bio-Plex calibration kit**	**Bio-Rad catalog #171-203001** **Bio-Rad catalog #171-203060**
Microtiter plate shaker IKA-Schuttler MTS-4 shaker for 4 microplates or Lab-Line Model 4625 Plate Shaker (or equivalent, capable of 300-1,100 rpm)	**IKA catalog #3208000** **VWR catalog #57019-600**
Filter plate vacuum apparatus Bio-Rad Aurum™ vacuum manifold **IMPORTANT:** The use of filter plate manifolds other than the one specified may result in diminished assay performance; see section 8 for instructions specific to this assay	**Bio-Rad catalog #732-6470**
Vortexer VWR brand mini-vortexer Scientific Instruments Vortex-Genie 2 mixer	**VWR catalog #58816-121** **VWR catalog #58815-234**
Reagent reservoir Corning, Inc. Costar 50 ml reagent reservoir 4870	**Bio-Rad catalog #224-4872**
Other materials Pipets and pipet tips, sterile distilled water, aluminum foil, absorbent paper towels, 1.5 ml microcentrifuge tubes, 15 ml culture tubes	

Section 6
Sample Preparation

This section provides instructions for preparing samples derived from serum, plasma, and tissue culture supernatant. For sample preparations not mentioned here, consult the publications listed in Bio-Rad bulletin 5297, available for download at **discover.bio-rad.com**

Serum and Plasma Samples

Note that for plasma samples, both EDTA plasma and citrate plasma are acceptable. Avoid using hemolyzed samples.

1. Collect and process the serum or plasma samples and assay immediately or freeze at –20ºC. Avoid repeated freeze-thaw cycles.

2. Centrifuge the samples at 13,200 rpm for 10 min at 4ºC to clear the samples of precipitate. Alternatively, carefully filter the samples with a 0.8/0.22 µm dual filter to prevent instrument clogging.

3. Immediately dilute 1 volume of sample with 3 volumes of sample diluent. Keep the samples on ice until ready for use.

Tissue Culture Supernatant Samples

1. Collect and process the tissue culture supernatant samples and assay immediately or freeze at –20ºC. Avoid repeated freeze thaw cycles.

2. If required, dilute the culture supernatant with culture medium. Serum-free culture medium should contain carrier protein (such as BSA) at a concentration of at least 0.5%. Keep the samples on ice until ready for use.

Section 7
Standard Preparation

Two vials of lyophilized angiogenesis standards are provided in each Bio-Plex Pro™ angiogenesis assay. However, only one vial is required per 96-well plate. The product insert provided with the assay lists the concentration of the reconstituted standard. This procedure will prepare enough standard to run each dilution in duplicate.

Reconstitute Standards

1. Gently tap the glass vial containing the lyophilized angiogenesis standard on a solid surface to ensure the pellet is at the bottom.
2. Reconstitute 1 vial of lyophilized standard with 500 µl of the standard diluent. Do not use assay buffer to dilute standards.
3. Gently vortex 1–3 sec and incubate on ice for 30 min.
4. Be consistent with the incubation time for optimal assay performance.

Prepare Standard Dilution Series

The angiogenesis concentrations specified for the 8-point standard dilution set have been selected for optimized curve fitting using the 5-parameter logistic (5PL) or 4-parameter logistic (4PL) regression in Bio-Plex Manager™ software. Results generated using dilution points other than those listed in this manual have not been optimized.

1. Label a set of 1.5 ml Eppendorf tubes as shown in the diagram on the next page.
2. Pipet the appropriate volume of standard diluent into the tubes (see diagram on next page). Use serum standard diluent for serum and plasma samples and culture medium for culture samples.

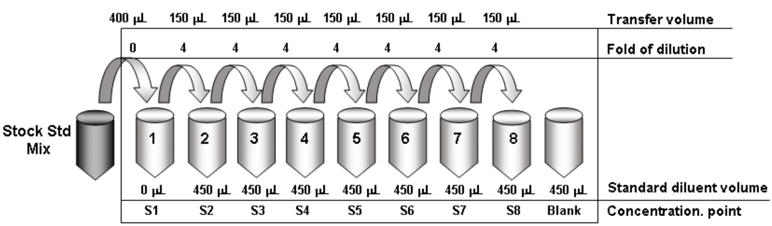
3. Add 400 µl of reconstituted standard to the first 1.5 ml tube with 0 µl of standard diluent. This is identified as S1 in the diagram below and in the product insert provided with assay.

4. Continue making serial dilutions of the standard as shown. After making each dilution, vortex gently and change the pipet tip after every transfer.

   NOTE: Running an additional two 0 pg/ml blanks is strongly recommended. The blank wells are useful for troubleshooting and determining LOD. Use 50 µl of the appropriate standard diluent as the blank sample. The 0 pg/ml points should be formatted as blanks, not as points in the curve, when using Bio-Plex Manager software. Formatting these wells as blanks automatically subtracts the background MFI values from the reading, and may result in negative MFI values in some wells. If negative MFI values are undesirable, format the 0 pg/ml wells as background controls.

5. Keep the standards on ice until ready for use. Standards should be used immediately and should not be frozen for future use.

Standard Dilution Series



Section 8
Control Preparation (Optional)

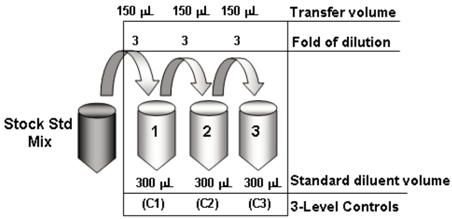
Two vials of lyophilized angiogenesis control are provided in each Bio-Plex Pro™ Angiogenesis assay. The preparation of high, medium, and low controls is optional to monitor plate-to-plate variations. This section provides instructions on how to reconstitute the lyophilized controls (refer to the product insert provided with the assay for reconstituted control values). The reconstituted control can then be further diluted to prepare any concentration of user-specified quality controls.

Reconstitute Angiogenesis Controls

To ensure optimal assay performance, angiogenesis controls should be prepared in a manner consistent as that used to prepare the angiogenesis standards.

1. Gently tap the glass vial containing the lyophilized angiogenesis control on a solid surface to ensure the pellet is at the bottom.
2. Reconstitute 1 vial of lyophilized control with 800 µl of the appropriate diluent. Do not use assay buffer to dilute controls. This is identified as stock in the product insert provided with the assay.
3. Gently vortex 1–3 sec and incubate on ice for 30 min. Be consistent with the incubation time to ensure optimal assay performance.
4. The reconstituted angiogenesis control should be further diluted to create the desired QC samples in the same diluents specified in the table below. Concentration of the reconstituted angiogenesis control can be found in the product insert provided with the assay.

Example of Control Dilution Series



Section 9
Assay Instructions

The following instructions apply to Bio-Plex Pro™ angiogenesis assays. All of the necessary components are provided premixed for ease of use.

Plan Experiment

1. Assign which wells of a 96-well plate will be used for each standard, control, and sample (see the example below).

2. Determine the total number of wells that will be used in the assay. Include a 25% excess (or add 2 wells for every 8 wells used) to ensure that enough diluted coupled beads, detection antibodies, and streptavidin-PE are prepared.

Example Plate

	1	2	3	4	5	6	7	8	9	10	11	12
A	1	1	B	B	5	5	13	13	21	21	29	29
B	2	2	1	1	6	6	14	14	22	22	30	30
C	3	3	2	2	7	7	15	15	23	23	31	31
D	4	4	3	3	8	8	16	16	24	24	32	32
E	5	5	1	1	9	9	17	17	25	25	33	33
F	6	6	2	2	10	10	18	18	26	26	34	34
G	7	7	3	3	11	11	19	19	27	27	35	35
H	8	8	4	4	12	12	20	20	28	28	36	36

Prepare Coupled Magnetic Beads

Protect the beads from light by covering the tubes with aluminum foil. Keep all tubes on ice until ready to use.

1. Vortex the coupled beads (25x) at medium speed for 15-20 sec.

2. Prepare a sufficient volume of coupled beads (1x) using assay buffer. Each well requires 2 µl of coupled beads (25x) adjusted to a final volume of 50 µl with assay buffer (refer to the example below).

Example Bead Calculations

# of Wells	25x Beads (µl)	Assay Buffer (µl)	Total Volume (µl)
96	240	5,760	6,000
48	120	2,880	3,000
32	80	1,920	2,000
24	60	1,440	1,500

Calibrate Vacuum Apparatus

The vacuum apparatus must be calibrated at the beginning of the assay to ensure an optimal bead yield. For more detailed instructions, refer to the Bio-Plex® suspension array system hardware instruction manual.

1. Prewet all the wells of a 96-well filter plate with 100 µl of assay buffer.

2. Place the filter plate on the vacuum apparatus and turn on the vacuum to the maximum level.

3. Press on the filter plate and note the time required to remove the buffer from the wells by vacuum filtration. The evacuation time should be 2–5 sec.

   If the evacuation time is <2 sec, the pressure is too high. Open the vacuum control valve slightly and repeat steps 1–3.

   If the evacuation time is >5 sec, the pressure is too low. Close the vacuum control valve slightly and repeat steps 1–3.

Assay Procedure

Bring all buffers to room temperature. Avoid bubbles when pipetting.

Assay Key – The following terms are repeated throughout the assay procedure. Refer to these detailed instructions when wash, incubate, and vacuum-filter are shown in bold.

Wash	Add 100µL of wash buffer to each well. Place the filter plate on a calibrated vacuum apparatus and remove the buffer by vacuum filtration. Blot the bottom of the filter plate with a clean paper towel. Repeat as specified.
	Tip: Thoroughly blot the bottom of the filter plate with a clean paper towel to prevent cross-contamination and plate leakage.
Incubate	Gently cover the filter plate with a new sheet of sealing tape. Place the filter plate on a microplate shaker and then cover with aluminum foil. Shake the filter plate at room temperature at 1,100 rpm for 30 sec, then at 300 rpm for the specified incubation time.
	Tip: Apply sealing tape gently on the filter plate (e.g. press down only on edges) to prevent positive pressure inside the wells that could lead to plate leaking during shaking. To avoid splashing of samples that may lead to cross-well contamination, slowly ramp up the shaker to the maximum speed before incubation. Cover plate with aluminum foil to prevent photobleaching.
Vacuum-filter	Place the filter plate on a calibrated vacuum apparatus and remove the buffer by vacuum filtration. Blot the bottom of the filter plate with a clean paper towel.
	Tip: Visually examine each well to ensure that buffers are completely drained from each well.

1. Equilibrate the diluted standards, samples, and controls at room temperature for 20 min prior to use.

2. Prewet and block the desired number of wells in a 96-well filter plate with 100 µl of assay buffer and **vacuum-filter**. If fewer than 96 wells are required, mark the plate to identify the unused wells for later use and cover the unused wells with sealing tape.

3. Vortex the coupled magnetic beads (1x) for 15–20 sec at medium speed. Add 50 µl to each well and immediately **vacuum-filter**.

4. **Wash** twice.

5. Gently vortex the diluted standards, controls, and samples for 1–3 sec. Add 50 µl of standard, control, or sample to each well, changing the pipet tip after every volume transfer. **Incubate** for 30 min.

6. While the samples are incubating, perform a 30 sec quick-spin centrifugation of the detection antibody (10x) prior to pipetting to collect the entire volume at the bottom of the vial.

7. Prepare a sufficient volume of detection antibodies (1x) using detection antibody diluent. Each well requires 2.5 µl of detection antibodies (10x) adjusted to a final volume of 25 µl with detection antibody diluent (refer to the example below).

Example Detection Antibody Calculations

# of Wells	10x Detection Antibody (µl)	Detection Antibody Diluent (µl)	Total Volume (µl)
96	300	2,700	3,000
48	150	1,350	1,500
32	100	900	1,000
24	75	675	750

8. After incubating the samples, slowly remove and discard the sealing tape, then **vacuum-filter**.

9. **Wash** 3 times.

10. Vortex the detection antibodies gently and add 25 µl to each well. **Incubate** for 30 min.

11. While the detection antibodies are incubating, perform a 30 sec quick-spin centrifugation of the streptavidin-PE (100x) prior to pipetting to collect the entire volume at the bottom of the vial.

12. Prepare a sufficient volume of streptavidin-PE (1x) using assay buffer. Each well requires 0.5 µl of streptavidin-PE (100x) adjusted to a final volume of 50 µl with assay buffer (refer to the example on the following page).

Example Streptavidin-PE Calculations

# of Wells	100x Streptavidin-PE (µl)	Assay Buffer (µl)	Total Volume (µl)
96	60	5,940	6,000
48	30	2,970	3,000
32	20	1,980	2,000
24	15	1,485	1,500

13. After the detection antibody incubation, slowly remove and discard the sealing tape, then **vacuum-filter**.

14. **Wash** 3 times.

15. Vortex the streptavidin-PE (1x) vigorously and add 50 µl to each well. **Incubate** for 10 min.

16. After the streptavidin-PE incubation, slowly remove and discard the sealing tape, then **vacuum-filter**.

17. **Wash** 3 times.

18. Add 125 µl of assay buffer to each well. Cover the filter plate with a new sheet of sealing tape. Shake the filter plate at room temperature at 1,100 rpm for 30 sec and remove the sealing tape before reading the plate.

Section 10
Data Acquisition

Bio-Plex Manager™ software is recommended for Bio-Plex Pro™ angiogenesis assays. Refer to the details in the appropriate section below for your xMAP system software package. For instructions using other xMAP system software packages, contact Bio-Rad technical support or your Bio-Rad field applications specialist.

NOTE: To minimize protocol setup, lot-specific Bio-Plex Pro angiogenesis assay protocols for Bio-Plex Manager version 4.0 and higher are available for download at **www.biorad.com/bio-plex/standards.**

Prepare System

1. Empty the waste bottle and fill the sheath fluid bottle before starting (if HTF not present). This will prevent fluidic system backup and potential data loss.

2. Turn on the reader, XY platform, and HTF (if included). Allow the system to warm up for 30 min.

3. Select Start up  and follow the instructions to prepare the reader to acquire data. If the system is idle for 4 hr without acquiring data, the lasers will automatically turn off. To reset the 4-hour countdown, select Warm up  and wait for the optics to reach operational temperature.

Calibrate With Low RP1 Target Value

Calibrate using Bio-Plex calibration beads and target values. Bio-Plex Pro angiogenesis assays are run using the low RP1 target value on the Cal 2 bottle. Refer to the details in the appropriate section below for your xMAP system software package.

Bio-Plex Manager Software version 5.0 and higher

These software versions require calibration only with the low RP1 target value. Daily calibration is recommended before acquiring data.

1. Select Calibrate and confirm that the default values for CAL1 and CAL2 are the same as the values on the Bio-Plex calibration bead labels. Use the Bio-Plex low RP1 target value.

2. Select OK and follow the instructions for CAL1 and CAL2 calibration.

Bio-Plex Manager Software version 4.0, 4.1 and 4.1.1
This software versions permit calibration at either low or high RP1 target values. The selection of which target value to use is assay dependent. For Bio-Plex Pro angiogenesis assays, calibrate using the low RP1 target value on the Cal 2 bottle. Re-calibration is necessary between assays requiring different RP1 target values. Daily calibration is recommended before acquiring data.

1. Select Calibrate and confirm that the default values for CAL1 and CAL2 are the same as the values on the Bio-Plex calibration bead labels. Use the Bio-Plex low RP1 target value.

2. Select OK and follow the instructions for CAL1 and CAL2 calibration.

Prepare Protocol

Lot-specific assay protocols are available for download at **www.bio-rad.com/bio-plex/standards**. If the desired protocol is not available for download, create a new protocol.

1. Open a new protocol by selecting File, then New from the main menu. Locate the steps at the left of the protocol menu.

2. Select Step 1 (Describe Protocol) and enter information about the assay.

3. Select Step 2 (Select Analytes) and choose the appropriate angiogenesis panel name, if available in the menu. If the panel name is not available, create a new panel:

a) Click the **Add Panel** button in the Select Analytes toolbar.

b) Enter a name for the new panel in the top field. Select the "Bio-Plex Pro" assay type from the pull down menu (only for version 5.0 or higher). Then click **Add** to add analytes.

c) Enter the bead region number of the first analyte in the Region field, and the analyte name in the Name field.

NOTE: The bead region number must be correct for proper detection of analytes. Using the product insert provided with the assay, confirm that this number is correct before proceeding.

d) Click **Add Continue** to add the analyte to the panel and continue adding more analytes. When you have entered your last analyte, click the Add button to add it to the list.

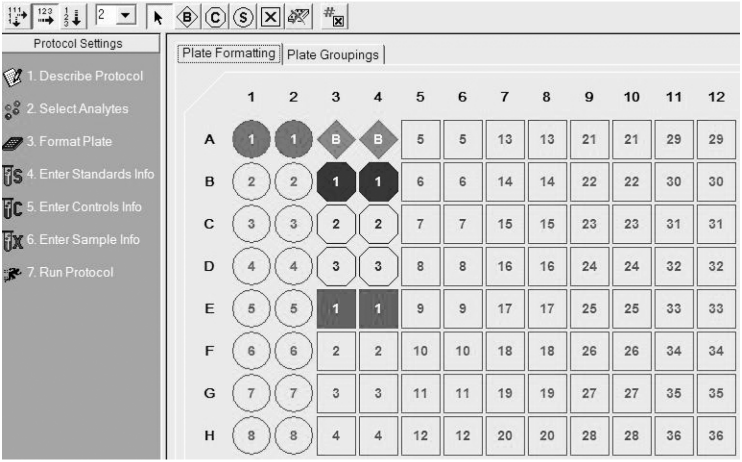
e) When you are finished creating the panel, click **OK** to save your changes and return to the Select Analytes window.

4. After selecting the panel, move analytes to the Selected view by choosing from analytes in the Available view using the Add button or Add All button. Note that the analytes will already be entered in the Selected view when using a downloaded protocol.

5. Select Step 3 (Format Plate) and click on the Plate Formatting tab. Select the number of replicates and the auto fill direction. Then click the S icon and drag the cursor over all the wells that contain standards. Then click on the B icon and drag the cursor over the wells that contain blanks. Repeat with C and X to identify all the wells that contain controls and samples.

   NOTE: If the protocol was downloaded from the website, the standards will already be added to the plate format. Make any necessary changes to the plate layout to match your plate requirements.

Plate Formatting Example



6. Select Step 4 (Enter Standards Info) to enter standards information. Skip to the next step if you are using a downloaded protocol from the website; this information will already be entered.

   a) Deselect the box labeled “same concentration values for all analytes”.

   b) Select the first analyte from the pull-down cell.

   c) Click on the Enter Automatically button and then select the most concentrated value (typically S1).

   d) Enter the value for the highest concentration. The information is included on the product insert provided with the assay.

   e) Enter the dilution factor (usually 4) and select Calculate. The concentrations for each replicate point of the standard will be populated for the selected analyte.

   f) Repeat steps 6b through 6e for each analyte in the assay.

7. Select Step 5 (Enter Controls Info) to enter controls information. This is where the concentration of the user-specified controls is entered into the protocol.

    a) If necessary, deselect the box for same concentration values for all analytes.

    b) Select an analyte from the pull down cell.

    c) Enter the description, concentration, and dilution information for each user-specified control.

    d) Repeat steps 7b and 7c for each additional analyte in the assay.

8. Select Step 6 (Enter Sample Info) and enter sample information. Remember to enter information for the appropriate dilution factor.

9. Select Step 7 (Run Protocol).

Acquire Data

1. Shake the assay plate at 1,100 rpm for 30 sec immediately before acquiring data.

2. Visually inspect the plate and ensure that the assay wells are filled with buffer prior to placing the plate in the Bio-Plex microplate platform.

3. Slowly remove the sealing tape and any plate cover before placing the plate in the reader.

4. Select Step 7 (Run Protocol) and refer to the details in the appropriate section below for your xMAP system software package:

    Bio-Plex Manager Software version 4.1, 4.1.1 and 5.0

    a) Specify data acquisition for **100 beads per region.**

    b) In Advanced Settings, confirm that the Bead Map is set to **25 region.**

    c) In Advanced Settings, set the sample size to **50 µ**l.

d) In Advanced Settings, confirm that the default DD gate values are set to **5000** (low) and **32000** (high).

e) Select Start, save the .rbx file, and begin data acquisition.

Bio-Plex Manager Software version 4.0

Note that this version does not include the 25 Bead Map. Hence, the 100 bead map is used.

a) Specify data acquisition for **100 beads per region**.

b) In Advanced Settings, set the sample size to **50 µl**.

c) In Advanced Settings, confirm that the default DD gate values are set to **5000** (low) and **32000** (high).

d) Select Start, save the .rbx file, and begin data acquisition.

5. If acquiring data from more than one plate, empty the waste bottle and refill the sheath bottle after each plate (if HTF not present). Select Wash Between Plates  and follow the instructions for fluidics maintenance. Then repeat the **Prepare Protocol** and **Acquire Data** steps.

   NOTE: Use the Wash Between Plates command after every plate run to reduce the possibility of clogging the instrument.

6. When data acquisition is complete, select Shut Down  and follow the instructions.

Reacquire Data

It is possible to aquire data from a well or plate a second time using the Rerun/Recovery mode located below Start in Step 7 (Run Protocol).

1. Check the wells where data will be acquired a second time. Any previous data will be overwritten.

2. Remove the buffer by vacuum filtration and add 100 µl of assay buffer to each well. Cover the filter plate with a new sheet of sealing tape.

3. Repeat **Acquire Data** steps 1-6 to acquire data a second time. The data acquired should be similar to the data acquired initially; however, the data acquisition time will be extended since fewer beads are present in each well.

Luminex® xPONENT® Software

Luminex xPONENT software does not include the 25-bead map. Hence, the 100-bead map is used. Refer to the detailed instructions in the xPONENT software manual.

1. Calibrate with xMAP **MagPlex** Classification Microspheres and xMAP Reporter Calibration Microspheres (sold by Luminex).

2. Verify with xMAP MagPlex Control Microspheres and xMAP Control Microspheres (sold by Luminex).

3. Create the angiogenesis panel protocol. Make sure to include the correct bead region numbers and target names (refer to the product insert).

3. The 100-region map should be selected.

4. Set the gate settings to **10,000** (low) to **22,500** (high).

Section 11
Troubleshooting Guides

This troubleshooting guide addresses problems that may be encountered with Bio-Plex Pro™ angiogenesis assays. If you experience any of the problems listed below, review the possible causes and solutions provided. This will assist you in resolving problems directly related to how the assay steps should be performed. Poor assay performance may also be due to the Bio-Plex® suspension array reader. To eliminate this possibility, we highly recommend use of the Bio-Plex validation kit. This kit will validate all the key functions of the array reader and assist the user in determining whether or not the array reader is functioning properly.

Possible Causes	Possible Solutions
High Inter-Assay CV	
Standards were not reconstituted consistently	Incubate the reconstituted standards for 30 min on ice. Always be consistent with the incubation time and temperature.
Reconstituted standards and diluted samples were not stored properly	Reconstituted standards and diluted samples should be prepared on ice as instructed. Prior to plating, the reconstituted standards and diluted samples should be equilibrated to room temperature.
High Intra-Assay CV	
Bottom of filter plate not dry	Dry the bottom of the filter plate with absorbent paper towel (preferably lint-free) to prevent cross-contamination.

Possible Causes	Possible Solutions
Pipetting technique	Pipet carefully and slowly when adding standards, samples, detection antibodies, and streptavidin-PE, especially when using a multichannel pipet. Use a calibrated pipet. Change pipet tip after every volume transfer.
Reagents and assay components were not equilibrated to room temperature prior to plating	All reagents and assay components should be equilibrated to room temperature prior to plating.
Contamination with wash buffer during wash steps	During the wash steps, be careful not to splash wash buffer from one well to another. Be sure that the wells are filtered completely and that no residual volume remains. Also, be sure that the microplate shaker setting is not too high. Reduce the microplate shaker speed to minimize splashing.
Slow pipeting samples and reagents across the plate	Sample pipeting across the entire plate should take less than 4 min. Reagent pipeting across the entire plate should take less than 1 min.

Possible Causes	Possible Solutions
Low Bead Count	
Miscalculation of bead dilution	Check your calculations and be careful to add the correct volumes.
Beads clumped in multiplex bead stock tube	Vortex for 15–20 sec at medium speed before aliquoting beads.
Vacuum on for too long when aspirating buffer from wells	Do not apply vacuum to the filter plate for longer than 10 sec after the buffer is completely drained from each well.
Did not shake filter plate enough before incubation steps and prior to reading	Shake the filter plate at 1,100 rpm for 30 sec before incubation steps and immediately before reading the plate.
Reader is clogged	Refer to the troubleshooting guide in the Bio-Plex hardware instruction manual.
Low Signal or Poor Sensitivity	
Standards reconstituted incorrectly	Follow the standard preparation instructions carefully (section 7).
Detection antibody or streptavidin-PE diluted incorrectly	Check your calculations and be careful to add the correct volumes.

Possible Causes	Possible Solutions
High Background Signal	
Incorrect buffer was used (for example, assay buffer used to dilute standards)	Use sample matrix or serum standard diluent to dilute the standards.
Spiked "0 pg/ml" wells by mistake	Be careful when spiking standards. Do not add any antigens in the 0 pg/ml (blank) point.
Streptavidin-PE incubated too long	Follow the procedure incubation time.
Poor Recovery	
Expired Bio-Plex reagents were used	Check that reagents have not expired. Use new or unexpired components.
Incorrect amounts of components were added	Check your calculations and be careful to add the correct volumes.
Microplate shaker set to an incorrect speed	Check the microplate shaker speed and use the recommended setting. Setting the speed too high may cause splashing and contamination. Use the recommended plate shaker.
Pipetting technique	Pipet carefully when adding standards, samples, detection antibodies, and streptavidin-PE, especially when using a multichannel pipet. Use a calibrated pipet. Change pipet tip after every volume transfer.

Section 12
Safety Considerations

Eye protection and gloves are recommended while using this product. Consult the MSDS for additional information.

The Bio-Plex Pro™ angiogenesis assays contain components of animal origin. This material should be handled as if capable of transmitting infectious agents. Please use universal precautions. These components should be handled at Biosafety Level 2 containment [US Government publication: Biosafety in Microbiological and Biomedical Laboratories (CDC, 1999)].

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*Including, but not limited to US patent 5,981,180; 6,046,807; 6,057,107.

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10010584 US/EG Rev A *Sig 1106*